Graphs of Sine and Cosine Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Amplitude, period, and midline**

For the function $y = 3\sin(2x)$, state:

a the amplitude $= 3$ **b** the period $= \pi$ **c** the range $[-3, 3]$

(a) $= 3$; (b) $= \pi$; (c) $[-3, 3]$

2 **[Foundational]** • **Target: Amplitude, period, and midline**

Consider the mathematical condition where for the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

a the amplitude $= 2$

b the period $= 4\pi$

c the midline $y = 1$

d the range $[-1, 3]$

(a) $= 2$; (b) $= 4\pi$; (c) $y = 1$; (d) $[-1, 3]$

3 **[Foundational]** • **Target: Amplitude, period, and midline**

For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

Phase shift: $\frac{\pi}{3}$ to the right. Period: 2π.

4 **[Foundational]** • **Target: Reading and writing equations from graphs**

The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.

Amplitude 2, period 2π. Equation: $y = 2\sin x$.

5 **[Foundational]** • **Target: Reading and writing equations from graphs**

Write an equation of a cosine function with amplitude 4, period π, reflected across the x-axis, with no phase or vertical shift.

$y = -4\cos(2x)$ (period $\frac{2\pi}{b} = \pi$ gives $b = 2$).

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6 [Foundational] • Target: Reading and writing equations from graphs

The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.

a State the maximum and minimum depths. Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.

b State the period of the tide cycle. Period $= \frac{2\pi}{\pi/6} = 12$ hours.

c Find the depth at $t = 3$. $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.

(a) Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.; (b) Period $= \frac{2\pi}{\pi/6} = 12$ hours.; (c) $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.

7 [Foundational] • Target: Graphing with transformations

The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.

Midline $y = 1$; maximum value 2; minimum value 0.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphing with transformations

Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.

Horizontal compression by a factor of 2 (period π), reflection across the x-axis, vertical stretch by a factor of 3, shift right $\frac{\pi}{4}$, and shift up 2.

9 [Intermediate] • Target: Graphing with transformations

State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

Amplitude 5; period $\frac{2\pi}{3}$; phase shift $\frac{\pi}{6}$ to the left; midline $y = -4$.

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10 [Intermediate] • Target: Modeling periodic phenomena

The average monthly temperature in a city varies between $8°\text{C}$ (in January, month 1) and $28°\text{C}$ (in July, month 7), following a sinusoidal pattern with a 12-month period.

a Find the amplitude and midline of the model. Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.

b Write a cosine model $T(m)$ for the temperature in month m, using the fact that the minimum occurs at $m = 1$. Since the minimum is at $m = 1$, use a negative cosine shifted:

$T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18$.

c Use your model to estimate the temperature in month 4 (April).

$T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18°\text{C}$.

(a) Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.; (b) Since the minimum is at $m = 1$, use a negative cosine shifted: $T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18$.; (c) $T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18°\text{C}$.

11 [Intermediate] • Target: Comparing sine and cosine graphs

Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.

The cosine graph looks identical to the sine graph shifted $\frac{\pi}{2}$ units to the LEFT. Shifting $\sin x$ left by $\frac{\pi}{2}$ means replacing x with $x + \frac{\pi}{2}$, giving $\sin\left(x + \frac{\pi}{2}\right) = \cos x$.

12 [Intermediate] • Target: Comparing sine and cosine graphs

Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.

Similarity: both have the same amplitude (2) and period (2π). Difference: the sine curve passes through the origin $(0, 0)$, while the cosine curve reaches its maximum at $x = 0$ — they are horizontally shifted versions of the same shape.

13 [Intermediate] • Target: Finding zeros and extrema from an equation

Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.

$2x = 0, \pi, 2\pi, 3\pi, 4\pi \Rightarrow x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

14 [Intermediate] • Target: Finding zeros and extrema from an equation

Find the x-value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Maximum value 3 occurs at $x = 0$ and $x = 2\pi$ (where $\cos x = 1$). Minimum value -3 occurs at $x = \pi$ (where $\cos x = -1$).

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Solving basic sinusoidal equations** • **Cross-Lesson: 7.02 Logarithms and Their Properties**

A Ferris wheel's height above ground is $h(t) = 15\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) + 17$ meters. Find the first time $t > 0$ (minutes) when the rider is at height 17 m and rising.

Height $= 17$ (the midline) when $\sin\left(\frac{\pi}{10}t - \frac{\pi}{2}\right) = 0$. The rider is rising through the midline when the argument equals 0: $\frac{\pi}{10}t - \frac{\pi}{2} = 0 \Rightarrow t = 5$ minutes.

16 **[Challenging]** • **Target: Reading and writing equations from graphs** • **Cross-Lesson: 7.03 Exponential and Logarithmic Equations**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

17 **[Challenging]** • **Target: Graphing with transformations** • **Cross-Lesson: 7.04 Applications: Growth and Decay**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

18 **[Challenging]** • **Target: Modeling periodic phenomena** • **Cross-Lesson: 8.01 Rational Functions and Asymptotes**

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2 - 1, 4 - 2, 1 - 3\rangle = \langle 1, 2, -2\rangle$. $\vec{AC} = \langle -1 - 1, 0 - 2, 5 - 3\rangle = \langle -2, -2, 2\rangle$. Step 2: Compute normal vector via cross product: $\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 2, -2; -2, -2, 2| = \mathbf{i}(4 - 4) - \mathbf{j}(2 - 4) + \mathbf{k}(-2 - (-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2\rangle$. Simplifying by scalar.

19 **[Challenging]** • **Target: Comparing sine and cosine graphs** • **Cross-Lesson: 8.02 Graphing Rational Functions**

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

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20 [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 8.03 Introduction to Limits

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2 + y^2 = r^2$, $x = r\cos\theta$, and $y = r\sin\theta$: $(r^2)^2 = 4(r^2\cos^2\theta - r^2\sin^2\theta) \Rightarrow r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$. Apply the double-angle identity $\cos^2\theta - \sin^2\theta = \cos(2\theta)$: $r^4 = 4r^2\cos(2\theta)$. Divide through by r^2 (for $r \neq 0$): $r^2 = 4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r = 2$ at.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Amplitude, period, and midline

Write a sinusoidal function with amplitude 5, period 4π, midline $y = -3$, using $\cos$.

a. $f(x) = 5\cos(1/2x) - 3$ ✓

b. $f(x) = 5\cos(1/2x) - 2$

c. $f(x) = 6\cos(1/2x) - 3$

d. $f(x) = 5\cos(1x) - 3$

Amplitude $= A = 5$; period $= 2\pi/B = 4\pi \Rightarrow B = 1/2$; midline $D = -3$. So $f(x) = 5\cos(1/2x) - 3$.

22 [Foundational] • Target: Reading and writing equations from graphs

The function $f(x) = -2\cos(\frac{\pi}{3}x) + 5$ has amplitude, period, and midline

a. **2, 6, y=5** ✓

b. 2, 3, y=5

c. –2, 6, y=5

d. 2, 6, y=0

Amplitude |A|=2, period = 2π/(π/3) = 6, midline y = 5.

23 [Foundational] • Target: Graphing with transformations

A sinusoidal function has period $\frac{2\pi}{3}$, midline $y = 3$, and minimum value -1. Find its amplitude.

a. 1

b. **4** ✓

c. 9

d. 2

Amplitude $=$ midline $-$ minimum $= 3 - (-1) = 4$.

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24 [Foundational] • Target: Modeling periodic phenomena

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?

a. 3

b. 4 ✓

c. 6

d. 8

We solve the iterated equation $f(f(f(x))) = 0$ layer by layer from the outside in. **Step 1: Outer Layer** Let $u = (f \circ f)(x)$. The equation becomes: $f(u) = 0 \Rightarrow \frac{1}{2}u^2 - 2 = 0 \Rightarrow u^2 = 4 \Rightarrow u = -2$ or $u = 2$ **Step 2: Middle Layer** Now set $v = f(x)$ and solve $f(v) = u$ for each value of u: - Case A: $f(v) = -2$ $\frac{1}{2}v^2 - 2 = -2 \Rightarrow \frac{1}{2}v^2 = 0 \Rightarrow v = 0$ Since -2 is the minimum value (vertex) of the parabola, this y.

25 [Foundational] • Target: Comparing sine and cosine graphs

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. **4** ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

26 [Foundational] • Target: Finding zeros and extrema from an equation

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. **4** ✓

c. 5

d. 16

The equation of a circle centered at the origin with radius r is given by $x^2 + y^2 = r^2$. In this problem, the equation of the circle is given as $x^2 + y^2 = 25$. This implies that the radius squared, r^2, is equal to 25. The circle passes through the point $P(3, y)$. Substituting the coordinates of point P into the equation of the circle, we get: $3^2 + y^2 = 25$. This simplifies to $9 + y^2 = 25$. To find the value of y^2, we subtract 9 from both sides: $y^2 = 25 - 9$, which gives $y^2 = 16$. Taking t.

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27 [Foundational] • Target: Solving basic sinusoidal equations

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$ ✓

The standard equation of a circle centered at the origin $(0, 0)$ with radius r is $x^2 + y^2 = r^2$. Since the circle passes through the point $(3, 4)$, we can substitute these coordinates into the equation to find r^2. $3^2 + 4^2 = r^2$. This gives $9 + 16 = r^2$, so $r^2 = 25$. Therefore, the equation of the circle is $x^2 + y^2 = 25$.

28 [Foundational] • Target: Amplitude, period, and midline

What is the radius of the circle with the equation $(x - 3)^2 + (y + 5)^2 = 36$?

a. **6** ✓

b. 36

c. 3

d. 5

The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius. In the given equation $(x - 3)^2 + (y + 5)^2 = 36$, we can see that $r^2 = 36$. To find the radius r, we take the square root of both sides: $r = \sqrt{36} = 6$. Therefore, the radius of the circle is 6.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Reading and writing equations from graphs

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$ ✓

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

Applying Polar Coordinates principles, Squaring $r = 1$: $r^2 = x^2 + y^2 = 1$. Therefore, the verified correct option is $x^2 + y^2 = 1$.

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30 [Intermediate] • Target: Graphing with transformations

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$ ✓

d. $\langle -8, 15 \rangle$

$|\mathbf{v}| = 17$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{-8}{17}, \frac{15}{17} \right\rangle$.

31 [Intermediate] • Target: Modeling periodic phenomena

A quantity starts at 100 and grows by 10% each period. Find its value after 2 periods.

a. **121** ✓

b. 100

c. 126

d. 120

Applying Exponential principles, $100(1 + \frac{1}{10})^2 = 121$. Therefore, the verified correct option is 121.

32 [Intermediate] • Target: Comparing sine and cosine graphs

A quantity starts at 100 and decays by 10% each period. Find its value after 2 periods.

a. **81** ✓

b. 100

c. 80

d. 86

Applying Exponential principles, $100(1 - \frac{1}{10})^2 = 81$. Therefore, the verified correct option is 81.

33 [Intermediate] • Target: Finding zeros and extrema from an equation

A quantity starts at 1600 and is divided by 2 every period. Find its value after 4 periods.

a. 1600

b. **100** ✓

c. 102

d. 6400

Applying Exponential principles, $\frac{1600}{2^4} = 100$. Therefore, the verified correct option is 100.

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34 [Intermediate] • Target: Solving basic sinusoidal equations

A quantity starts at 50 and multiplies by 3 every period. Find its value after 3 periods.

a. 1353

b. **1350** ✓

c. 150

d. 50

Applying Exponential principles, $50 \times 3^3 = 1350$. Therefore, the verified correct option is 1350.

35 [Intermediate] • Target: Amplitude, period, and midline

A population starts at 100 and doubles every period. Find its value after 3 periods.

a. **800** ✓

b. 300

c. 100

d. 802

Applying Exponential principles, $100 \times 2^3 = 800$. Therefore, the verified correct option is 800.

36 [Intermediate] • Target: Reading and writing equations from graphs

A population starts at 3 and is multiplied by 3 every period. Find its value after 3 periods.

a. 3

b. 84

c. 9

d. **81** ✓

Applying Exponential principles, $3 \times 3^3 = 81$. Therefore, the verified correct option is 81.

37 [Intermediate] • Target: Graphing with transformations

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?

a. $y = \pm\frac{3}{2}x$ ✓

b. $y = \pm\frac{2}{3}x$

c. $y = \pm 6x$

d. $y = \pm\frac{5}{2}x$

Asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x = \pm\frac{3}{2}x$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Modeling periodic phenomena • Cross-Lesson: 8.01 Rational Functions and Asymptotes

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. **1** ✓

$r = 2a\cos\theta$ is the circle $(x-a)^2 + y^2 = a^2$ with $2a = 2 \Rightarrow a = 1$; radius $= 1$.

39 [Challenging] • Target: Comparing sine and cosine graphs • Cross-Lesson: 8.02 Graphing Rational Functions

A matrix has 1 rows and 3 columns. What are its dimensions?

a. **1×3** ✓

b. 4

c. 3

d. 3×1

Matrix dimensions are written rows×columns: 1×3.

40 [Challenging] • Target: Finding zeros and extrema from an equation • Cross-Lesson: 8.03 Introduction to Limits

Use Cramer's Rule to solve: $5x - 3y = 14$, $-3x + 4y = -4$.

a. $x = 4, y = 3$

b. $x = 4, y = 2$ ✓

c. $x = 5, y = 2$

d. $x = 2, y = 4$

$D = 11$. $D_x = 44$, so $x = D_x/D = 4$. $D_y = 22$, so $y = D_y/D = 2$.

41 [Challenging] • Target: Solving basic sinusoidal equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Eliminate the parameter: $x = t$, $y = -1t + 5$.

a. $x = -1y + 5$

b. $y = -1x$

c. $y = -1x + 5$ ✓

d. $y = 1x + 5$

Substitute $t = x$ into the equation for y: $y = -1x + 5$.

42 [Challenging] • Target: Amplitude, period, and midline • Cross-Lesson: 9.02 Geometric Sequences and Series

Find the determinant of $(-1, 2;\ -6, 4)$.

a. **8** ✓

b. -16

c. 9

d. 7

$\det = ad - bc = (-1)(4) - (2)(-6) = -4 - (-12) = 8$.

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43 [Challenging] • Target: Reading and writing equations from graphs • Cross-Lesson: 9.03 The Binomial Theorem

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3 ✓

b. -1

c. -2

d. -13

We consider the piecewise definition of $f(x) = |x^2 - 6x + 5| = |(x - 1)(x - 5)|$: xx = 2 \pm \sqrt{-1 - k} $liestrictlyin$(1, 5), $contributing2solutions$. $-$ Atk = -2, $therootsare$x = 1andx = 3. $Since$x = 1 $isanendpoint$, $only$x = 3$liesin$(1, 5), $contributing1solution$. $-$ $Fork$ \in (-10, -2), $onlythelargerroot2$ + \sqrt{-1 - k}$liesin$(1, 5), $contributing1solution$. 2. $**$ $Outersections$(x \le 1orx \ge 5) : $**$ $Setting$ x^2 - 6x + 5 = 2x + kx The roots are $x = 4 \pm \sqrt{11 + k}$. - For.

44 [Challenging] • Target: Graphing with transformations • Cross-Lesson: 1.01 Functions and Function Notation

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

a. **c = 3, k = 4** ✓

b. c = 5, k = 3

c. c = 3, k = 3

d. c = 5, k = 4

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

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45 [Challenging] • Target: Modeling periodic phenomena • Cross-Lesson: 1.02 Domain and Range

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

a. $A = 3, B = 4$ ✓

b. $A = 3, B = 2$

c. $A = 6, B = 4$

d. $A = 6, B = 2$

The amplitude A is half the difference between the maximum and minimum values: $A = \frac{5-(-1)}{2} = \frac{6}{2} = 3$. Since $A > 0$ is assumed, $A = 3$. The vertical shift D is the average of the maximum and minimum values: $D = \frac{5+(-1)}{2} = \frac{4}{2} = 2$. The period of the function is given by $T = \frac{2\pi}{|B|}$. We are given that the period is $\frac{\pi}{2}$, so $\frac{2\pi}{B} = \frac{\pi}{2}$ (since $B > 0$). Solving for B, we get $B = \frac{2\pi}{\pi/2} = 4$. Thus, $A = 3$ and $B = 4$.

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